1. Project Planning & Management

Project Title : Traffic Accident Analysis

Project Overview :-

The Traffic Accident Analysis project aims to analyze road accident data obtained from the Kaggle dataset. By using data analysis tools such as Python, SQL, Excel, Power BI, and Tableau, the project identifies the main causes of traffic accidents, discovers accident patterns and trends, and presents the results through interactive dashboards. The insights generated can help decision-makers improve road safety and reduce accident rates.

Objectives :-

. Analyze traffic accident data to identify the main causes of accidents.

. Discover patterns and trends related to traffic accidents.

. Visualize data using interactive dashboards.

. Support decision-making through data-driven insights.

. Provide recommendations to reduce traffic accidents and improve road safety.

Scope ,The project includes:-

. Collecting the dataset from Kaggle.

. Cleaning and preprocessing the data.

. Analyzing the data using SQL and Python.

. Creating dashboards using Power BI, Tableau, and Excel.

. Presenting findings, insights, and recommendations.

The project does not include:

Machine Learning prediction models.

Developing a web or mobile application.

Project Plan :

Project Duration: 2.5 Months (10 Weeks)

Duration

. Project Planning Week 1

. Data Collection Week 2

. Data Cleaning & Preprocessing Weeks 3–4

. Data Analysis Weeks 5–6

. Dashboard Development Weeks 7–8

. Testing & Documentation Week 9

. Final Presentation Week 10

Task Assignment & Roles :-

Team Member Responsibility

Mina Gamal Fahmy : Excel Analysis & Dashboard

Sara Wael Mohamed : Power BI Dashboard Development

Mariam Mohamed : Python Data Cleaning & Analysis

Menna Ali : Tableau Dashboard Development

Ahmed Ali : SQL Data Extraction & Queries

Salah Osama Project Support, Troubleshooting & Integration

Risk Assessment & Mitigation Plan :-

Risk Mitigation

Missing or incomplete data : Clean the dataset and handle missing values before analysis.

Delay in completing tasks : Follow the project schedule and hold regular team meetings.

Technical or software issues : Troubleshoot problems and use backup solutions if needed.

Integration issues between tools : Test outputs regularly and ensure compatibility.

Data quality issues : Validate and verify the dataset before analysis.

KPIs (Key Performance Indicators) :-

Based on the project dashboard:

Total Accidents: 152M

Total Casualties: 235M

Average Casualties per Accident: 19.07K

Average Vehicles Involved per Accident: 25.13K

Sum of total accident ;151683856

2. Literature Review:-

Literature Review :-

Previous studies have used traffic accident data to identify the main factors related to road accidents. Factors such as weather, road conditions, vehicle type, time, and accident severity can help in understanding accident patterns.

Feedback & Evaluation :-

Previous analyses helped identify important accident factors, but some of them focused on specific factors and did not provide a complete view of the data. Also, some results were not presented in a simple way for users to understand.

Suggested Improvements :-

Our project will provide a more complete analysis of the available accident data. We will use data analysis and visualization to identify important patterns and present the results through clear dashboards and simple insights.

3. Requirements Gathering :-

Stakeholder Analysis

Stakeholder Role

Traffic Authorities : Use the results to understand accident causes and improve road safety.

Data Analysts : Clean, analyze, and visualize the accident data.

Decision Makers : Use the results to support decisions related to road safety.

Researchers : Use the analysis to study accident patterns and causes.

General Public : Benefit from recommendations that can help improve road safety.

Functional Requirements :-

. Import the traffic accident dataset from Kaggle.

. Clean and preprocess the data.

. Analyze accident data using SQL and Python.

. Create interactive dashboards using Power BI, Tableau, and Excel.

. Display KPIs such as Total Accidents, Total Casualties, and Accident Severity.

. Filter data by area, day, weather, and accident severity.

. Generate reports and visualizations.

Non-functional Requirements

. The system should provide accurate analysis results.

. Dashboards should be easy to understand and use

. The project should process the dataset efficiently.

. Data should remain consistent and reliable.

. The project should be well-organized and easy to maintain.

4. System Analysis & Design

Problem Statement & Objectives

Problem Statement :-

Road traffic accidents lead to significant casualties, injuries, and property damage. Identifying the primary causes, understanding high-risk conditions, and evaluating driver demographics are essential challenges in reducing accident rates and improving overall road safety.

Objectives:

. Identify the primary causes of road traffic accidents using historical data analysis.

.Analyse the correlation between environmental conditions (such as weather, lighting, and road surface state) and accident severity.

. Evaluate driver demographics (including age brackets and driving experience) and their direct involvement in accidents.

. Provide actionable, data-driven recommendations and safety strategies to minimize accident rates.

Database Design & Data Modeling (ERD)

Dataset Overview :-

Project Title: Road Traffic Accidents (RTA) Analysis

Dataset Scope: The dataset contains 12,316 records across 27 categorical and numerical variables covering driver profiles, vehicle types, environmental conditions, and accident severities.

ERD Entities & Keys :

Accident Entity (Central Entity)

Primary Key (PK): Accident_ID

Foreign Keys (FK): Driver_ID, Vehicle_ID, Environment_ID

Attributes: Time, Day_of_week, Accident_severity

Driver Profile Entity

Primary Key (PK): Driver_ID

Foreign Keys (FK): None

Attributes: Age_band_of_driver, Sex_of_driver, Driving_experience

Environment Entity

Primary Key (PK): Environment_ID

Foreign Keys (FK): None

Attributes: Area_accident_occured, Road_surface_conditions, Weather_conditions

Vehicle Entity

Primary Key (PK): Vehicle_ID

Foreign Keys (FK): None

Attributes: Type_of_vehicle

Logical Schema (Relational Structure) :-

Accident : Accident_ID (PK), Driver_ID, Vehicle_ID, Environment_ID, Time, Day_of_week, Accident_severity

Driver : Driver_ID (PK), Age_band_of_driver, Sex_of_driver, Driving_experience

Environment : Environment_ID PK, Area_accident_occured, Road_surface_conditions, Weather_conditions

Vehicle : Vehicle_ID (PK), Type_of_vehicle

Physical Schema

table name : accident

Column Name	Data Type	Column Name	Data Type
Accident_ID	INT	Time	TIME
Driver_ID	INT	Day_of_week	VARCHAR
Vehicle_ID	INT	Accident_severity	VARCHAR
Environment_ID	INT		

table name : driver

Driver_ID	INT
Age_band_of_driver	VARCHAR
Sex_of_driver	VARCHAR
Driving_experience	VARCHAR

<u>table name :</u> Environment

Environment_ID INT

Road_surface_conditions VARCHAR

Weather_conditions VARCHAR

<u>table name :</u> Vehicle

Vehicle_ID INT

Type_of_vehicle VARCHAR

5. Implementation :

Source Code :

- **Excel Data Processing:** Applying advanced Excel functions and formulas (such as XLOOKUP, INDEX/MATCH, SUMIFS) to clean and prepare data for analysis.
- **Power BI DAX & M Code:** Writing measures and calculated columns using DAX, and using M Code in Power Query for data transformation and cleaning (ETL process).

Coding Standards & Naming Conventions

- **Naming Standards:** Assigning clear names to tables and dynamic measures to enhance readability (e.g., Total_Sales or Fact_Sales).
- **Organization:** Grouping measures inside dedicated "Measure Tables" and applying uniform formatting for numbers and currencies.

Modular Code & Reusability

- **Reusability:** Designing flexible data models (Star Schema / Snowflake Schema) and setting up Power Query templates that update automatically as source data changes.
- **Components:** Creating reusable control parameters and slicers across different report pages.

Security & Error Handling

- **Data Security:** Implementing **Row-Level Security (RLS)** to control data access permissions based on user roles.
- **Error Handling:** Managing missing values (Nulls) and code errors in DAX and Power Query rules using functions like ISBLANK and DIVIDE to prevent division-by-zero errors.

GitHub Repository

- **Repository Structure:** Uploading source project files (.pbix, .xlsx, .csv) along with documentation files inside the repository.
- **Version Control:** Documenting updates and modifications with clear commits to track changes over time.

6. Testing & Quality Assurance

Test Cases & Test Plan

- **Data Validation:** Conducting comparison tests on final results and totals between Excel and Power BI to ensure calculation accuracy.

- **Performance & Interactivity Testing:** Verifying report response times when using slicers and testing interactive features (e.g., Drill-through and Cross-filtering).

Bug Reports

- **Bug Logging:** Documenting any issues encountered during analysis (e.g., slow DAX queries, duplicate records, or broken table relationships) alongside the steps taken to resolve and prevent them.

7. Final Presentation & Reports

Technical Documentation

- **Technical Guide:** A comprehensive manual detailing the data model relationships, data sources used, transformations applied in Power Query, and all DAX formulas implemented.

Project Presentation

- **Presentation Deck:** An interactive slide deck highlighting project objectives, key insights, and visual analytics extracted from the Power BI Dashboard, along with actionable business recommendations.

User Manual

- **User Guide:** A simple guide explaining to the client/end-user how to navigate the Power BI Dashboard, interact with filters, switch between pages, and export reports.